

DATA STRUCTURES (LAB)

LAB TASK 02



By

MISBAH KHALID

FA25-BCS-005

Submitted to,

Mr. BILAL BUKHARI

September 17, 2026

Department of Computer Sciences
COMSATS University Islamabad,
Attock Campus

TASK 01

CODE:

```
#include <iostream>
using namespace std;

struct Node
{
    int data;
    Node* next;
};

// Insert node at end
void insert(Node*& first, int value)
{
    Node* newNode = new Node;
    newNode->data = value;
    newNode->next = NULL;

    if (first == NULL)
    {
        first = newNode;
    }
    else
    {
        Node* p = first;

        while (p->next != NULL)
        {
            p = p->next;
        }

        p->next = newNode;
    }
}

// Display normally
void display(Node* first)
{
    Node* p = first;

    while (p != NULL)
    {
```

```

        cout << p->data << " ";
        p = p->next;
    }
}

// Reverse using loop
void reverseLoop(Node* first)
{
    int arr[100];
    int i = 0;

    Node* p = first;

    while (p != NULL)
    {
        arr[i] = p->data;
        i++;
        p = p->next;
    }

    for (i = i - 1; i >= 0; i--)
    {
        cout << arr[i] << " ";
    }
}

// Reverse using recursion
void reverseRecursive(Node* p)
{
    if (p == NULL)
        return;

    reverseRecursive(p->next);

    cout << p->data << " ";
}

int main()
{
    Node* first = NULL;

    insert(first, 10);
    insert(first, 20);

```

```

insert(first, 30);
insert(first, 40);
insert(first, 50);

cout << "Original List: ";
display(first);

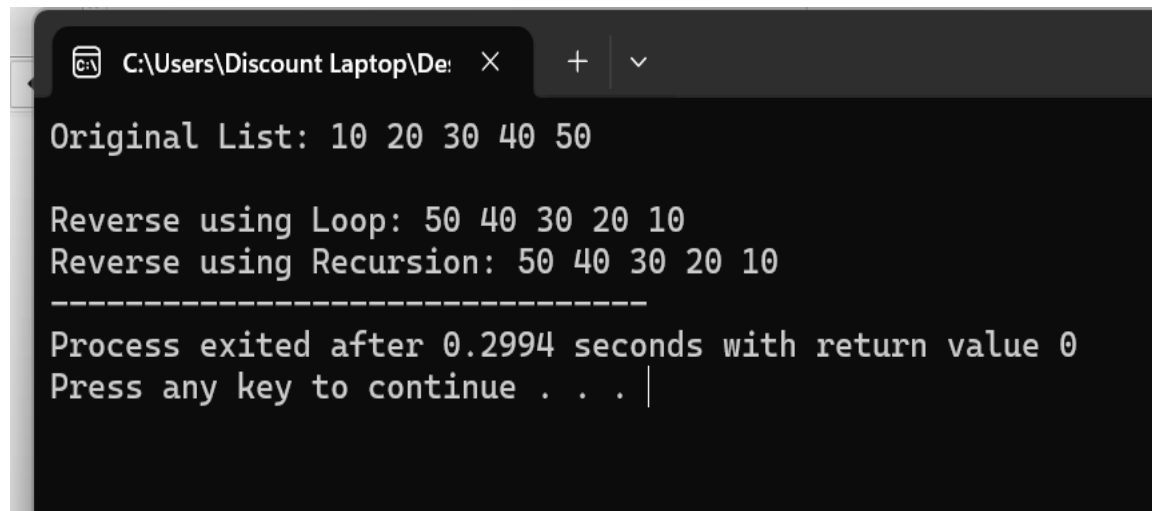
cout << "\n\nReverse using Loop: ";
reverseLoop(first);

cout << "\nReverse using Recursion: ";
reverseRecursive(first);

return 0;
}

```

OUTPUT:



```

C:\Users\Discount Laptop\De:
Original List: 10 20 30 40 50

Reverse using Loop: 50 40 30 20 10
Reverse using Recursion: 50 40 30 20 10
-----
Process exited after 0.2994 seconds with return value 0
Press any key to continue . . . |

```

TASK 02

CODE:

```

#include <iostream>

using namespace std;

struct Node
{

```

```
int data;

Node* next;

};


// Insert at end

void insert(Node*& first, int value)
{
    Node* newNode = new Node;
    newNode->data = value;
    newNode->next = NULL;

    if (first == NULL)
    {
        first = newNode;
    }
    else
    {
        Node* p = first;

        while (p->next != NULL)
        {
            p = p->next;
        }
    }
}
```

```
    p->next = newNode;
}
}
```

// Merge two lists into third list

```
void mergeLists(Node* first1, Node* first2, Node*& first3)
{
    Node* p = first1;

    while (p != NULL)
    {
        insert(first3, p->data);
        p = p->next;
    }

    p = first2;

    while (p != NULL)
    {
        insert(first3, p->data);
        p = p->next;
    }
}
```

```
// Display list

void display(Node* first)
{
    Node* p = first;

    while (p != NULL)
    {
        cout << p->data << " ";
        p = p->next;
    }
}
```

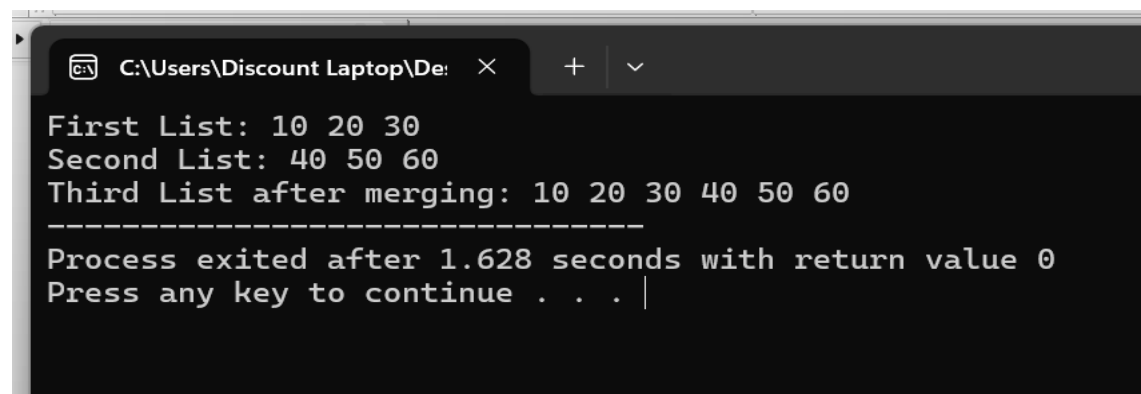
```
int main()
{
    Node* first1 = NULL;
    Node* first2 = NULL;
    Node* first3 = NULL;

    // First list
    insert(first1, 10);
    insert(first1, 20);
    insert(first1, 30);

    // Second list
```

```
insert(first2, 40);  
insert(first2, 50);  
insert(first2, 60);  
  
cout << "First List: ";  
display(first1);  
  
cout << "\nSecond List: ";  
display(first2);  
  
// Merge  
mergeLists(first1, first2, first3);  
  
cout << "\nThird List after merging: ";  
display(first3);  
  
return 0;  
}
```

OUTPUT:



```
C:\Users\Discount Laptop\De:  ×  +  ▾  
First List: 10 20 30  
Second List: 40 50 60  
Third List after merging: 10 20 30 40 50 60  
-----  
Process exited after 1.628 seconds with return value 0  
Press any key to continue . . . |
```

TASK 03

CODE:

```
#include <iostream>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    int data;
```

```
    Node* next;
```

```
};
```

```
// Insert at end
```

```
void insert(Node*& first, int value)
```

```
{
```

```
    Node* newNode = new Node;
```

```
    newNode->data = value;
```

```
    newNode->next = NULL;
```

```
    if (first == NULL)
```

```
    {
```

```
        first = newNode;
```

```
    }
```

```
    else
```

```
    {
```

```
        Node* p = first;
```

```

while (p->next != NULL)
{
    p = p->next;
}

p->next = newNode;
}
}

// Find multiple occurrences
void findOccurrences(Node* first, int value)
{
    Node* p = first;
    int position = 1;
    int count = 0;

    while (p != NULL)
    {
        if (p->data == value)
        {
            cout << "Found at position: " << position << endl;
            count++;
        }
    }
}

```

```

        p = p->next;
        position++;
    }

    if (count == 0)
    {
        cout << "Value not found." << endl;
    }
    else
    {
        cout << "Total occurrences: " << count << endl;
    }
}

// Display list
void display(Node* first)
{
    Node* p = first;

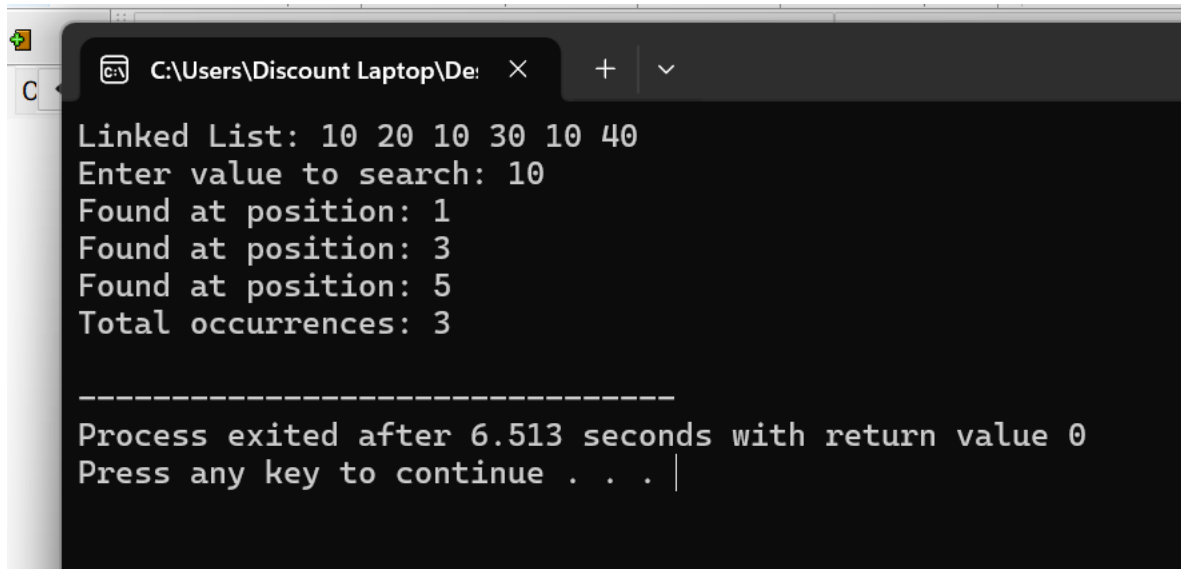
    while (p != NULL)
    {
        cout << p->data << " ";
        p = p->next;
    }
}

```

```
    }  
}  
  
int main()  
{  
    Node* first = NULL;  
  
    insert(first, 10);  
    insert(first, 20);  
    insert(first, 10);  
    insert(first, 30);  
    insert(first, 10);  
    insert(first, 40);  
  
    cout << "Linked List: ";  
    display(first);  
  
    int value;  
  
    cout << "\nEnter value to search: ";  
    cin >> value;  
  
    findOccurrences(first, value);
```

```
    return 0;  
}
```

OUTPUT:



```
C:\Users\Discount Laptop\De:  X  +  v  
C  
Linked List: 10 20 10 30 10 40  
Enter value to search: 10  
Found at position: 1  
Found at position: 3  
Found at position: 5  
Total occurrences: 3  
  
-----  
Process exited after 6.513 seconds with return value 0  
Press any key to continue . . . |
```